

Funciones CES

Martín Cicowiez
(CEDLAS-UNLP)

FCE-UNLP
Marzo 23, 2021

CES functions

- On the production side in most applied CGE models, CES functions are preferred to Cobb-Douglas functions. The reason: With the help of additional elasticity data, it is possible to control the ease with which activities can substitute one factor for another, including distinctions between activities and the possibility of testing alternative assumptions.
 - Recall: Cobb-Douglas functions impose an elasticity of substitution of 1.

The cost minimization problem

$$QA = \varphi \left(\delta_L \cdot QF_L^{-\rho} + \delta_K \cdot QF_K^{-\rho} \right)^{\frac{-1}{\rho}} \quad \delta_L + \delta_K = 1 \quad (1)$$

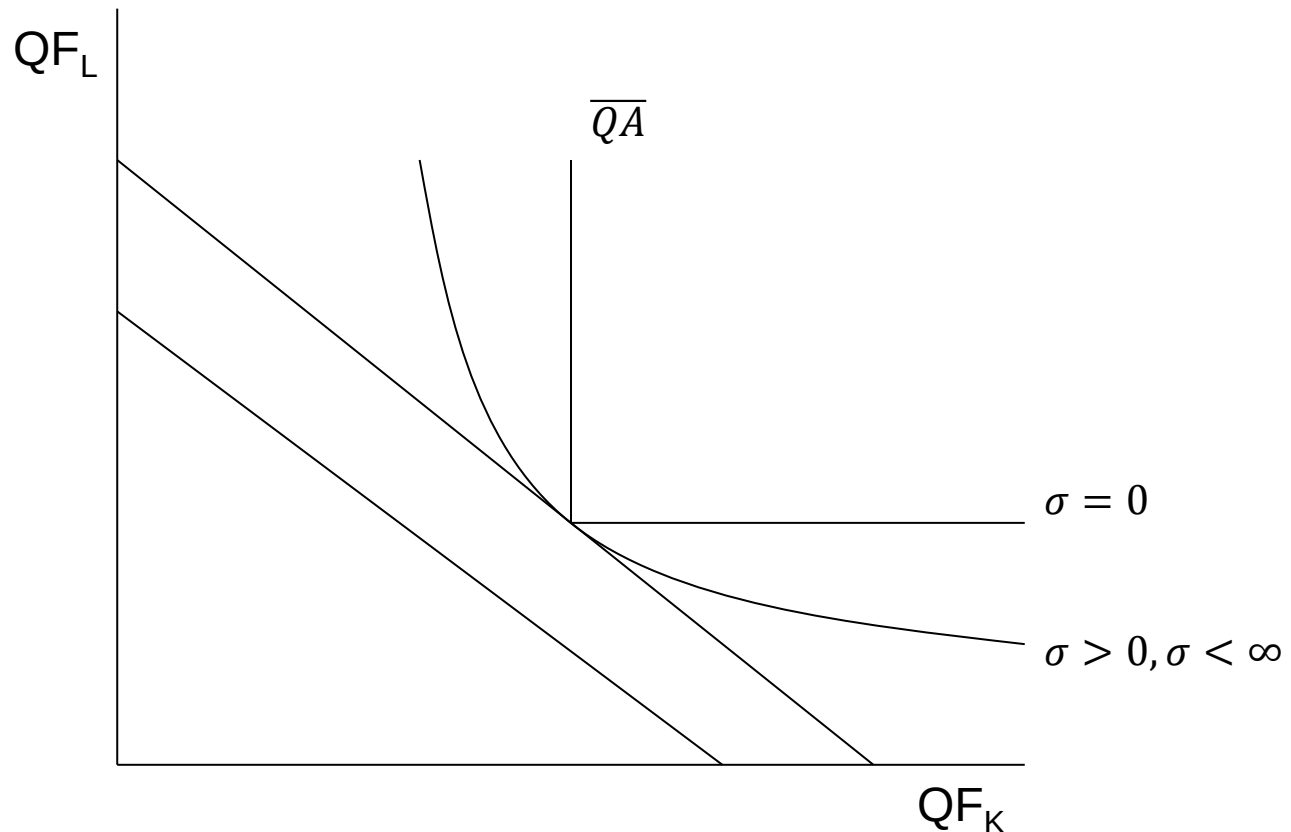
where QA is a composite volume (e.g., production), QF_f are inputs, φ is a scale parameter, δ_f are share parameter, and ρ is related to the elasticity of substitution, σ . The producer / consumer problem is to minimize the cost, C , of obtaining QA (which is fixed), taking (1) as a constraint:

$$\min C = \overline{WF}_L \cdot QF_L + \overline{WF}_K \cdot QF_K$$

$$\text{s. t. } \overline{QA} = \varphi \left(\delta_L \cdot QF_L^{\frac{\sigma-1}{\sigma}} + \delta_K \cdot QF_K^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} \quad \sigma = \frac{1}{1 + \rho}$$

The cost minimization problem – cont.

$$\min Z = \overline{WF}_L \cdot QF_L + \overline{WF}_K \cdot QF_K + \lambda \left[\overline{QA} - \varphi \left(\delta_L \cdot QF_L^{\frac{\sigma-1}{\sigma}} + \delta_K \cdot QF_K^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} \right]$$



The elasticity of substitution

$$\sigma = \frac{\frac{d(QF_L/QF_K)}{QF_L/QF_K}}{\frac{d(WF_K/WF_L)}{WF_K/WF_L}}$$

- Interpretation: for the CES function, the elasticity of the ratio between the quantities of the input pair L-K with respect to the K-L price ratio is a constant (σ).
- Example: for $\sigma = 0.1$, if the price of input K increases by 10% relative to the price of input L, the input ratio L / K input increases by 1%.
- Note: the CES function is not defined for $\sigma = 0$ or 1, since some terms in the CES function would be divided by zero.

First-order conditions

First-order conditions (FOCs; the derivatives of the Lagrangian with respect to its endogenous variables):

$$\text{(FOC1)} \quad \frac{\partial Z}{\partial QF_L} = \lambda \cdot \varphi \cdot \frac{\sigma}{\sigma-1} \left(\delta_L \cdot QF_L^{\frac{\sigma-1}{\sigma}} + \delta_K \cdot QF_K^{\frac{\sigma-1}{\sigma}} \right)^{\frac{1}{\sigma-1}} \delta_L \cdot \frac{\sigma-1}{\sigma} \cdot QF_L^{\frac{-1}{\sigma}} = \overline{WF}_L$$

$$\text{(FOC2)} \quad \frac{\partial Z}{\partial QF_K} = \lambda \cdot \varphi \cdot \frac{\sigma}{\sigma-1} \left(\delta_L \cdot QF_L^{\frac{\sigma-1}{\sigma}} + \delta_K \cdot QF_K^{\frac{\sigma-1}{\sigma}} \right)^{\frac{1}{\sigma-1}} \delta_K \cdot \frac{\sigma-1}{\sigma} \cdot QF_K^{\frac{-1}{\sigma}} = \overline{WF}_K$$

$$\text{(FOC3)} \quad \frac{\partial Z}{\partial \lambda} = \overline{QA} - \varphi \left(\delta_L \cdot QF_L^{\frac{\sigma-1}{\sigma}} + \delta_K \cdot QF_K^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} = 0$$

FOC – cont.

Manipulating FOC1 to derive the cost-minimizing demand function for input 1:

$$QF_L = \left[\frac{\lambda \cdot \varphi \left(\delta_L \cdot QF_L^{\frac{\sigma-1}{\sigma}} + \delta_K \cdot QF_K^{\frac{\sigma-1}{\sigma}} \right)^{\frac{1}{\sigma-1}}}{\overline{WF}_L} \delta_L \right]^{\sigma}$$

Manipulating FOC3:

$$\frac{\overline{QA}}{\varphi} = \left(\delta_L \cdot QF_L^{\frac{\sigma-1}{\sigma}} + \delta_K \cdot QF_K^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}$$

FOC – cont.

Substituting FOC3 into the previous equation to get the demand function for input 1:

$$QF_L = \left(\frac{\lambda}{\overline{WF_L}} \right)^\sigma \delta_L^\sigma \cdot \varphi^{\sigma-1} \cdot \overline{QA}$$

Parallel result for input 2:

$$QF_K = \left(\frac{\lambda}{\overline{WF_K}} \right)^\sigma \delta_K^\sigma \cdot \varphi^{\sigma-1} \cdot \overline{QA}$$

Defining calibrated CES parameters

Defining the share and efficiency parameters (δ 's and φ) of the CES input demand functions from σ , SAM values, price normalization, FOCs, and the fact that

$\delta_L + \delta_K = 1$. Starting with δ_L , from FOC1 and FOC2:

$$(1) \frac{\delta_L \cdot QF_L^{\frac{-1}{\sigma}}}{\delta_K \cdot QF_K^{\frac{-1}{\sigma}}} = \frac{WF_L}{WF_K}$$

$$(4) \delta_L \cdot QF_K^{\frac{1}{\sigma}} \cdot WF_K = (1 - \delta_L) QF_L^{\frac{1}{\sigma}} \cdot WF_L$$

$$(2) \delta_L \cdot QF_L^{\frac{-1}{\sigma}} \cdot WF_K = \delta_K \cdot QF_K^{\frac{-1}{\sigma}} \cdot WF_L$$

$$(5) \delta_L \left(QF_K^{\frac{1}{\sigma}} \cdot WF_K + QF_L^{\frac{1}{\sigma}} \cdot WF_L \right) = QF_L^{\frac{1}{\sigma}} \cdot WF_L$$

$$(3) \delta_L \cdot QF_K^{\frac{1}{\sigma}} \cdot WF_K = \delta_K \cdot QF_L^{\frac{1}{\sigma}} \cdot WF_L$$

$$(6) \delta_L = \frac{WF_L \cdot QF_L^{\frac{1}{\sigma}}}{WF_K \cdot QF_K^{\frac{1}{\sigma}} + WF_L \cdot QF_L^{\frac{1}{\sigma}}}$$

Defining calibrated CES parameters – cont.

Parallel result for δ_K :

$$\delta_K = \frac{WF_K \cdot QF_K^{\frac{1}{\sigma}}}{WF_K \cdot QF_K^{\frac{1}{\sigma}} + WF_L \cdot QF_L^{\frac{1}{\sigma}}}$$

The scale parameter, φ , is defined from the production function – this is possible given that, at this stage, all other values in this function are known:

$$\varphi = \frac{\overline{QA}}{\left(\delta_L \cdot QF_L^{\frac{\sigma-1}{\sigma}} + \delta_K \cdot QF_K^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}}$$

Special cases for input demand and production functions

$\sigma = 0 \Rightarrow$ Leontief

Input demand:

$$QF_f = \delta_f \cdot QA$$

Production function:

$$QA = \min\left(\frac{QF_L}{\delta_L}, \frac{QF_K}{\delta_K}\right)$$

δ_f = quantity of input f per unit of output

$\sigma = 1 \Rightarrow$ Cobb-Douglas

Input demand:

$$QF_f = \frac{\delta_f \cdot \lambda \cdot QA}{WF_f}$$

Production function:

$$QA = \varphi \cdot QF_L^{\delta_L} \cdot QF_K^{\delta_K}$$